03 Quadratic Equation

1. INTRODUCTION:

The algebraic expression of the form $ax^2 + bx + c, a \neq 0$ is called a quadratic expression, because the highest order term in it is of second degree. Quadratic equation means, $ax^2 + bx + c = 0$. In general, whenever one says zeroes of the expression $ax^2 + bx + c$, it implies roots of the equation $ax^2 + bx + c = 0$, unless specified otherwise.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

2. SOLUTION OF QUADRATIC EQUATION & RELATION BETWEEN ROOTS & CO-EFFICIENT:

(a) The general form of quadratic equation is $ax^2 + bx + c = 0, a \neq 0$.

The roots can be found in following manner:

$$a\left(x^2+\frac{b}{a}x+\frac{c}{a}\right)=0 \Rightarrow \left(x+\frac{b}{2a}\right)^2+\frac{c}{a}-\frac{b^2}{4a^2}=0$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}-\frac{c}{a} \Rightarrow \boxed{x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}}$$

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This expression can be directly used to find the two roots of a quadratic equation.

(b) The expression $b^2 - 4ac = D$ is called the discriminant of the quadratic equation.

(c) If $\alpha \& \beta$ are the roots of the quadratic equation $ax^2 + bx + c = 0$, then:

(i) $\alpha+\beta = -b/a$ (ii) $\alpha\beta = c/a$ (iii) $|\alpha-\beta|=\frac{\sqrt{D}}{|a|}$

(d) A quadratic equation whose roots are $\alpha \& \beta$ is $(x - \alpha)(x - \beta) = 0$ i.e. $x^2 - (\alpha+\beta)x + \alpha\beta = 0$ i.e. x^2 – (sum of roots) x + product of roots = 0.

Illustration 1

If α and β are the roots of $ax^2 + bx + c = 0$, find the equation whose roots are $\alpha + 2$ and $\beta + 2$.

Solution:

Replacing x by $x - 2$ in the given equation, the required equation is

$\Rightarrow a(x-2)^2 + b(x-2) + c = 0$ i.e., $ax^2 - (4a - b)x + (4a - 2b + c) = 0$.

Illustration 2

The coefficient of x in the quadratic equation $x^2 + px + q = 0$ was taken as 17 in place of 13, its roots were found to be –2 and –15. Find the roots of the original equation.

Solution:

Here $q = (-2) \times (-15) = 30$, correct value of $p = 13$. Hence original equation is

$\Rightarrow x^2 + 13x + 30 = 0$ as $(x + 10)(x + 3) = 0$

$\therefore$ roots are $-10, -3$

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Illustration 3

If α, β are the roots of a quadratic equation, $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is -

(A) $x^2 + 4x + 1 = 0$ (B) $x^2 - 4x + 4 = 0$

(C) $x^2 - 4x - 1 = 0$ (D) $x^2 + 2x + 3 = 0$

Ans. (B)

Solution:

Since α, β are the roots of equation $x^2 - 3x + 5 = 0$

So $\alpha^2 - 3\alpha + 5 = 0$

$\Rightarrow \beta^2 - 3\beta + 5 = 0$

$\therefore \alpha^2 - 3\alpha = -5$

$\Rightarrow \beta^2 - 3\beta = -5$

Putting in $(\alpha^2 - 3\alpha + 7)$ & $(\beta^2 - 3\beta + 7)$ ...(i)

$\Rightarrow -5 + 7, -5 + 7$

$\therefore$ 2 and 2 are the roots.

$\therefore$ The required equation is $x^2 - 4x + 4 = 0$.

Illustration 4

If α and β are the roots of $ax^2 + bx + c = 0$, find the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$.

Solution:

If $\alpha + \beta$ are roots of $ax^2 + bx + c = 0$

then, $a\alpha^2 + b\alpha + c = 0 \Rightarrow a\alpha + b = -\frac{c}{\alpha}$

same as $a\beta + b = -\frac{c}{\beta}$

$\Rightarrow (a\alpha + b)^{-2} = (\alpha\beta + b)^{-2} = \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2}$

$= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{c^2} = \frac{(-b/a)^2 - 2(c/a)}{c^2} = \frac{b^2 - 2ac}{a^2c^2}$

Illustration 5

If the roots of the equation $x^2 + 2ax + b = 0$ are real and distinct and they differ by atmost $2m$, then b lies in the interval

(A) $(a^2 - m^2, a^2)$ (B) $[a^2 - m^2, a^2)$ (C) $(a^2, a^2 + m^2)$ (D) none of these

Ans. (B)

Solution:

$x^2 + 2ax + b = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$\Rightarrow 0 < |\alpha - \beta| \leq 2m$

$\Rightarrow 0 < \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \leq 2m$

$\Rightarrow 0 < 4a^2 - 4b \leq 4m^2$

$\Rightarrow a^2 - m^2 \leq b < a^2$

$\Rightarrow b \in [a^2 - m^2, a^2)$

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Illustration 6

The set of possible values of λ for which $x^2 - (\lambda^2 - 5\lambda + 5)x + (2\lambda^2 - 3\lambda - 4) = 0$ has roots, whose sum and product are both less than 1, is

(A) $\left(-1, \frac{5}{2}\right)$ (B) $(1, 4)$ (C) $\left[1, \frac{5}{2}\right]$ (D) $\left(1, \frac{5}{2}\right)$

Ans. (D)

Solution:

Sum of roots < 1

$\Rightarrow \lambda^2 - 5\lambda + 5 < 1$

$\Rightarrow (\lambda - 1)(\lambda - 4) < 0$

$\Rightarrow 1 < \lambda < 4$ ...(1)

$\Rightarrow$ Product of roots < 1

$\Rightarrow 2\lambda^2 - 3\lambda - 4 < 1$

$\Rightarrow 2\lambda^2 - 3\lambda - 5 < 0$

$\Rightarrow (2\lambda - 5)(\lambda + 1) < 0$

$\Rightarrow -1 < \lambda < \frac{5}{2}$ ...(2)

(1) & (2)

$\Rightarrow 1 < \lambda < \frac{5}{2}$.

3. NATURE OF ROOTS:

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(a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then; $x = \frac{-b \pm \sqrt{D}}{2a}$

(i) $D > 0$ roots are real & distinct (unequal).

(ii) $D = 0$ roots are real & coincident (equal)

(iii) $D < 0$ roots are imaginary.

(iv) If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa. $(p, q \in \mathbb{R})$

(b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then;

(i) If D is a perfect square, then roots are rational.

(ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is an irrational) then other root will be $p - \sqrt{q}$

Illustration 7

For what values of 'm' the equation $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has equal roots.

Solution:

Given equation is $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ ...(i)

Let D be the discriminant of equation (i).

Roots of equation (i) will be equal if $D = 0$.

$\Rightarrow 4(1 + 3m)^2 - 4(1 + m)(1 + 8m) = 0$

$\Rightarrow 4(1 + 9m^2 + 6m - 1 - 9m - 8m^2) = 0$

$\Rightarrow m^2 - 3m = 0$ or $m(m - 3) = 0$

$\therefore m = 0, 3.$

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Illustration 8

Find all the integral values of 'a' for which the quadratic equation $(x-a)(x-10)+1=0$ has integral roots.

Solution:

Here the equation is $x^2-(a+10)x+10a+1=0$. Since integral roots will always be rational it means D should be a perfect square.

From (i) $D=a^2-20a+96$.

$\Rightarrow D=(a-10)^2-4$

$\Rightarrow 4=(a-10)^2-D$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when

$\Rightarrow (a-10)^2=4$ and $D=0$.

$\Rightarrow (a-10)=\pm 2$

$\Rightarrow a=12,8$

Illustration 9

Show that the expression $x^2+2(a+b+c)x+3(bc+ca+ab)$ will be a perfect square if $a=b=c$.

Solution:

Given quadratic expression will be a perfect square if the discriminant of its corresponding equation is zero.

i.e. $4(a+b+c)^2-4.3(bc+ca+ab)=0$

$\Rightarrow (a+b+c)^2-3(bc+ca+ab)=0$

$\Rightarrow \frac{1}{2}\left((a-b)^2+(b-c)^2+(c-a)^2\right)=0$

which is possible only when $a=b=c$.

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Illustration 10

If the coefficient of the quadratic equation are rational & the coefficient of x^2 is 1, then find the equation one of whose roots is $\sqrt{2}-1$.

Solution:

Irrational roots always occur in conjugate pairs.

Hence if one root is $(-1+\sqrt{2})$, the other root will be $(-1-\sqrt{2})$. Equation is

$(x-(-1+\sqrt{2}))(x-(-1-\sqrt{2}))=0$

$\Rightarrow x^2+2x-1=0$

Illustration 11

Find the set of all real values of λ such that the root of the equation

$x^2+2(a+b+c)x+3\lambda(ab+bc+ca)=0$

are always real for any choice of a,b,c (where a,b,c represents sides of scalene triangle).

(A) $\left(-\infty,\frac{4}{3}\right)$ (B) $\left(\frac{4}{3},\infty\right)$ (C) $\left(\frac{1}{3},\frac{5}{3}\right)$ (D) $\left(\frac{4}{3},\frac{5}{3}\right)$

Ans. (A)

Solution:

We, know that $a+b>c, b+c>a$ and $c+a>b$

$\Rightarrow c-a<b, a-b<c, b-c<a$

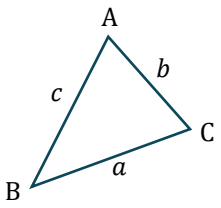
squaring on both sides and adding $(c-a)^2+(a-b)^2+(b-c)^2<a^2+b^2+c^2$

$a^2+b^2+c^2-2(ab+bc+ca)<0$

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$\Rightarrow (a + b + c)^2 - 4(ab + bc + ca) < 0$

$\Rightarrow \frac{(a+b+c)^2}{ab+bc+ca} < 4$ ...(i)



Now roots of equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then $D \geq 0$

$\Rightarrow 4(a + b + c)^2 - 4.3\lambda(ab + bc + ca) \geq 0$

$\Rightarrow \frac{(a+b+c)^2}{ab+bc+ca} \geq 3\lambda$

So $3\lambda \leq \frac{(a+b+c)^2}{ab+bc+ca} < 4 \Rightarrow \lambda < \frac{4}{3}$

Illustration 12

If coefficients of biquadratic equation are all distinct and belong to the set $\{-9, -5, 3, 4, 7\}$, then equation has

(A) atleast two real roots

(B) four real roots, two are conjugate surds and other two are also conjugate surds

(C) four imaginary roots

(D) None of these

Ans. (A)

Solution:

Let biquadratic is $ax^4 + bx^3 + cx^2 + dx + e = 0$

$\Rightarrow a + b + c + d + e = 0$ as $a, b, c, d, e \in \{-9, -5, 3, 4, 7\}$

Hence $x = 1$ is a root. So real root will be atleast two.

Illustration 13

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Let $p, q, r, s \in R, x^2 + px + q = 0, x^2 + rx + s = 0$ such that $2(q + s) = pr$ then

(A) atleast one of the equation have real roots.

(B) either both equations have imaginary roots or both equations have real roots.

(C) one of equations have real roots and other equation have imaginary roots

(D) atleast one of the equations have imaginary roots.

Ans. (A)

Solution:

$\Rightarrow x^2 + px + q = 0 \Rightarrow D_1 = p^2 - 4q$ ...(1)

$\Rightarrow x^2 + rx + s = 0 \Rightarrow D_2 = r^2 - 4s$ ...(2)

$\Rightarrow D_1 + D_2 = p^2 + r^2 - 4(q + s)$ $[\because pr = 2(q + s)]$

$\Rightarrow (p - r)^2 > 0$

Since $D_1 + D_2$ is +ve, so atleast one of the equations has real roots.

4. SYMMETRIC EXPRESSIONS OF α AND β:

Let α and β be the roots of the equation $ax^2 + bx + c = 0$ such that $f(\alpha, \beta) = f(\beta, \alpha)$ then $f(\alpha, \beta)$ denotes symmetric expression of the roots.

e.g. $f(\alpha,\beta) = \alpha^2\beta + \alpha\beta^2$

It is to be noted that every symmetric expression in α,β can be expressed in terms of two symmetric expression $\alpha + \beta$ and $\alpha\beta$

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5. TRANSFORMATION OF THE EQUATION:

Let $ax^2 + bx + c = 0$ be a quadratic equation with two roots α and β. If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α and β, then this equation can be found by finding α in terms of y. Now as α satisfies given equation, put this α in terms of y directly in the equation.

$y = f(\alpha)$

By transformation, $\alpha = g(y)$

$a(g(y))^2 + b(g(y)) + c = 0$

This is the required equation in y.

Illustration 14

If the roots of $ax^2 + bx + c = 0$ are α and β, then find the equation whose roots are:

(a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$ (b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ (c) α^2, β^2

Solution:

(a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$

put, $y = \frac{-2}{\alpha} \Rightarrow \alpha = \frac{-2}{y}$

$\Rightarrow a\left(-\frac{2}{y}\right)^2 + b\left(\frac{-2}{y}\right) + c = 0 \Rightarrow cy^2 - 2by + 4a = 0$

Required equation is $cx^2 - 2bx + 4a = 0$

(b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

put, $y = \frac{\alpha}{\alpha+1} \Rightarrow \alpha = \frac{y}{1-y}$

$\Rightarrow a\left(\frac{y}{1-y}\right)^2 + b\left(\frac{y}{1-y}\right) + c = 0$

$\Rightarrow (a + c - b)y^2 + (-2c + b)y + c = 0$

Required equation is $(a + c - b)x^2 + (b - 2c)x + c = 0$

(c) α^2, β^2 put $y = \alpha^2 \Rightarrow \alpha = \sqrt{y}$

$\Rightarrow ay + b\sqrt{y} + c = 0$

$\Rightarrow b^2y = a^2y^2 + c^2 + 2acy$

$\Rightarrow a^2y^2 + (2ac - b^2)y + c^2 = 0$

Required equation is $a^2x^2 + (2ac - b^2)x + c^2 = 0$

Illustration 15

If the roots of $ax^3 + bx^2 + cx + d = 0$ are α, β, γ then find equation whose roots are $\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}$.

Solution:

Put $y = \frac{1}{\alpha\beta} = \frac{\gamma}{\alpha\beta\gamma} = -\frac{a\gamma}{d}$ $(\alpha\beta\gamma = -\frac{d}{a})$

Put $x = -\frac{dy}{a}$

$\Rightarrow a\left(-\frac{dy}{a}\right)^3 + b\left(-\frac{dy}{a}\right)^2 + c\left(-\frac{dy}{a}\right) + d = 0$

Required equation is $d^2x^3 - bdx^2 + acx - a^2 = 0$

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6. EQUATION VS IDENTITY:

An equation which is true for every value of the variable within the domain is called an identity, for example: $5(a-3) = 5a - 15$, $(a+b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in R$.

Note: An equation of degree ≤ 2 cannot have three or more roots & if it has, it becomes an identity.

If $ax^2 + bx + c = 0$ is an identity $\Leftrightarrow a = b = c = 0$

Illustration 16

If the equation $(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2)x + (\lambda^2 - 4) = 0$ has more than two roots, then find the value of λ ?

Ans. ($\lambda = 2$)

Solution:

As the equation has more than two roots so it becomes an identity. Hence

$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, 3$

And $\lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1, 2$

and $\lambda^2 - 4 = 0 \Rightarrow \lambda = 2, -2$

So $\lambda = 2$

7. COMMON ROOTS:

Consider two quadratic equations, $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$.

(i) If two quadratic equations have both roots common, then the equations are identical and their co-efficient are in proportion.

i.e. $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(ii) If one root is common, then the common root 'α' will be:

$$\Rightarrow \alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$$

Hence the condition for common root is:

$$\Rightarrow (c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

Note:

If $f(x) = 0$ & $g(x) = 0$ are two polynomial equation having some common root(s) then those common root(s) is/are also the root(s) of $h(x) \equiv a\,f(x) + bg(x) = 0$.

Illustration 17

Determine 'a' such that $x^2 - 11x + a$ and $x^2 - 14x + 2a$ may have a common factor.

Solution:

Let $x - \alpha$ be a common factor of $x^2 - 11x + a$ and $x^2 - 14x + 2a$.

Then $x = \alpha$ will satisfy the equations $x^2 - 11x + a = 0$ and $x^2 - 14x + 2a = 0$.

$\therefore\ \alpha^2 - 11\alpha + a = 0$ and $\alpha^2 - 14\alpha + 2a = 0$

Solving (i) and (ii) by cross multiplication method, we get $a = 0, 24$.

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Illustration 18

If $x^2 - a\,x + b = 0$ and $x^2 - p\,x + q = 0$ have a root in common and the second equation has equal roots, show that $b + q = \frac{ap}{2}$.

Solution:

Given equations are : $x^2 - a\,x + b = 0$ ...(i)

and $x^2 - p\,x + q = 0.$ ...(ii)

Let α be the common root. Then roots of equation (ii) will be α and α. Let β be the other root of equation (i). Thus, roots of equation (i) are α, β and those of equation (ii) are α, α.

Now

$\Rightarrow \alpha + \beta = a$ ...(iii)

$\Rightarrow \alpha\beta = b$ ...(iv)

$\Rightarrow 2\alpha = p$ ...(v)

$\Rightarrow \alpha^2 = q$ ...(vi)

L.H.S. = $b + q = \alpha\beta + \alpha^2 = \alpha(\alpha + \beta)$ ...(vii)

and R.H.S. = $\frac{ap}{2} = \frac{(\alpha+\beta)2\alpha}{2} = \alpha(\alpha + \beta)$ ...(viii)

from (vii) and (viii), L.H.S. = R.H.S.

Illustration 19

If $a, b, c \in R$ and equations $ax^2 + bx + c = 0$ and $x^2 + 2x + 9 = 0$ have a common root, show that $a : b : c = 1 : 2 : 9$.

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Solution:

Given equations are: $x^2 + 2x + 9 = 0$ ...(i)

and $ax^2 + bx + c = 0$ ...(ii)

Clearly roots of equation (i) are imaginary since equation (i) and (ii) have a common root, therefore common root must be imaginary and hence both roots will be common.

Therefore equations (i) and (ii) are identical

$\Rightarrow \frac{a}{1} = \frac{b}{2} = \frac{c}{9}$

$\Rightarrow\ a : b : c = 1 : 2 : 9$

Illustration 20

Find p and q such that $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots in common.

Solution:

$a_1 = p, b_1 = 5, c_1 = 2$ and $a_2 = 3, b_2 = 10, c_2 = q$

We know that:

$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{p}{3} = \frac{5}{10} = \frac{2}{q}$

$\Rightarrow p = \frac{3}{2}\,; q = 4$

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Illustration 21

Find the conditions on a, b, c and d such that equations

$2ax^3 + bx^2 + cx + d = 0$ and $2ax^2 + 3bx + 4c = 0$ have a common root.

Solution:

Let 'α' be a common root of the given equations,

then $2a\alpha^3 + b\alpha^2 + c\alpha + d = 0$ ...(1)

and $2a\alpha^2 + 3b\alpha + 4c = 0$ ...(2)

Multiply (2) by α and then subtract (1) from it, to get

$2b\alpha^2 + 3c\alpha - d = 0$ ...(3)

Now (2) and (3) are quadratic having a common root α, so

$\Rightarrow \frac{\alpha^2}{-3bd-12c^2} = \frac{\alpha}{8bc+2ad} = \frac{1}{6ac-6b^2}$, $\alpha^2 = \frac{bd+4c^2}{2b^2-2ac}$, $\alpha = \frac{4bc+ad}{3ac-3b^2}$

Eliminating α from these two equations, we get

$(4bc + ad)^2 = \frac{9}{2}(bd + 4c^2)(b^2 - ac)$, which is the required condition.

Illustrations 22

If α, β are the roots of $x^2 + px + q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$,

evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

in terms of p, q, r and s. Deduce the condition that the equations have a common root.

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Solution:

α, β are the roots of $x^2 + px + q = 0$

$\Rightarrow \alpha + \beta = -p, \alpha\beta = q$ ...(1)

and γ, δ are the roots of $x^2 + rx + s = 0$

$\Rightarrow \gamma + \delta = -r, \gamma\delta = s$ ...(2)

Now, $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

$\Rightarrow [\alpha^2 - \alpha(\gamma + \delta) + \gamma\delta][\beta^2 - \beta(\gamma + \delta) + \gamma\delta]$

$\Rightarrow (\alpha^2 + r\alpha + s)(\beta^2 + r\beta + s)$

$\Rightarrow \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2$

$\Rightarrow \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s((\alpha + \beta)^2 - 2\alpha\beta)) + sr(\alpha + \beta) + s^2$

$\Rightarrow q^2 - pqr + r^2q + s(p^2 - 2q) + sr(-p) + s^2$

$\Rightarrow (q - s)^2 - rpq + r^2q + sp^2 - prs$

$\Rightarrow (q - s)^2 - rq(p - r) + sp(p - r)$

$\Rightarrow (q - s)^2 + (p - r)(sp - rq)$

For a common root (Let $\alpha = \gamma$ or $\beta = \delta$) ...(3)

Then $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = 0$ ...(4)

from (3) and (4), we get

$(q - s)^2 + (p - r)(sp - rq) = 0$

$\Rightarrow (q - s)^2 = (p - r)(rq - sp)$, which is the required condition.

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8. RATIONAL INEQUALITIES:

Method of Intervals (Wavy Curve Method):

Let $g(x)=\left(\frac{(x-b_1)^{k_1}(x-b_2)^{k_2}---(x-b_n)^{k_n}}{(x-a_1)^{r_1}(x-a_2)^{r_2}---(x-a_n)^{r_n}}\right)$ ...(i)

Where $k_1,k_2,.....k_n$ and $r_1,r_2.....r_n \in \mathbb{N}$ and $b_1,b_2.....b_n$ and $a_1,a_2.....a_n$ are real numbers.

Then to solve the inequality following steps are taken.

Steps:

Points where numerator becomes zero are called zeros or roots of the function and where denominator becomes zero are called poles of the function.

(i) First we find the zeros and poles of the function.

(ii) Then we mark all the zeros and poles on the real line and put a vertical bar there dividing the real line in many intervals.

(iii) Determine sign of the function in any of the interval and then alternates the sign in the neighbouring interval if the poles or zeros dividing the two intervals has appeared odd number of times otherwise retain the sign.

(iv) Thus, we consider all the intervals. The solution of the $g(x)>0$ is the union of the intervals in which we have put the plus sign and the solution of $g(x)<0$ is the union of all intervals in which we have put the minus sign.

Illustration 23

Find x such that $3x^2-7x+6<0$

Solution:

$f(x)=3x^2-7x+6$

$\Rightarrow D=49-72<0$ and $a>0$

$\Rightarrow 3x^2-7x+6>0\ \forall x\in R \Rightarrow x\in\phi$

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Illustration 24

Find x such that $2x^2+4x+9>0$

Solution:

$f(x)=2x^2+4x+9$

$\Rightarrow a>0$ and $D<0$

$\Rightarrow 2x^2+4x+9>0\ \forall\ x\in\mathbb{R}$

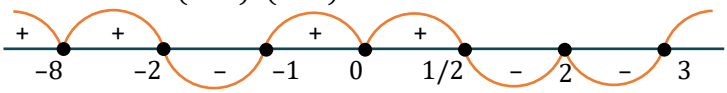
$\Rightarrow x\in(-\infty,\infty)$

Illustration 25

Solve the inequality if $f(x)=\frac{(x-2)^{10}(x+1)^3\left(x-\frac{1}{2}\right)^5(x+8)^2}{x^{24}(x-3)^3(x+2)^5}$ is >0 or <0.

Solution:

Let $f(x)=\frac{(x-2)^{10}(x+1)^3\left(x-\frac{1}{2}\right)^5(x+8)^2}{x^{24}(x-3)^3(x+2)^5}$ the poles and zeros are $0,3,-2,-1,\frac{1}{2},-8,2$



If $f(x)>0$, then $x\in(-\infty,-8)\cup(-8,-2)\cup(-1,0)\cup\left(0,\frac{1}{2}\right)\cup(3,\infty)$

and if $f(x)<0$, then $x\in(-2,-1)\cup\left(\frac{1}{2},2\right)\cup(2,3)$.

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Illustration 26

Number of positive integral values of x satisfying the inequality

$$\frac{(x-4)^{2013}.(x+8)^{2014}(x+1)}{x^{2016}(x-2)^3.(x+3)^5.(x-6)(x+9)^{2012}} \le 0 \text{ is}$$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. (D)

Solution:

Poles and zeros are $4, -8, -1, 0, 2, -3, 6, -9$



$x \in (-\infty, -9) \cup (-9, -3) \cup [-1, 0) \cup (0, 2) \cup [4, 6)$

so +ve integral solutions are $x = 1, 4, 5$

Illustration 27

Number of non-negative integral values of x satisfying the inequality

$$\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \ge 0 \text{ is}$$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. (D)

Solution:

$$\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \ge 0$$

$$\frac{2x+2-x^2+x-1}{x^3+1} - \frac{(2x-1)}{x^3+1} \ge 0 \Rightarrow \frac{3x+1-x^2-2x+1}{x^3+1} \ge 0$$

$$\Rightarrow \frac{-x^2+x+2}{x^3+1} \ge 0 \Rightarrow \frac{x^2-x-2}{x^3+1} \le 0$$

$$\Rightarrow \frac{(x+1)(x-2)}{(x+1)(x^2-x+1)} \le 0, x \ne -1 \Rightarrow \frac{(x-2)}{(x^2-x+1)} \le 0$$



$x \in (-\infty, -1) \cup (-1, 2]$

required value of $x = \{0, 1, 2\}$.

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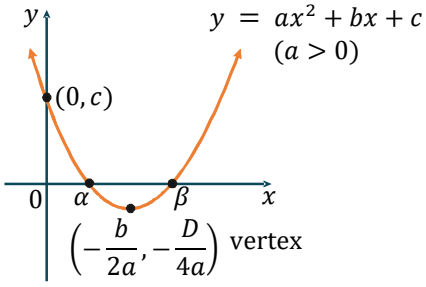
9. GRAPH OF QUADRATIC EXPRESSION:

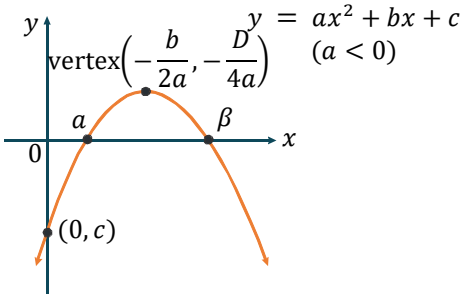
$y = f(x) = ax^2 + bx + c$ **or** $\left(y + \frac{D}{4a}\right) = a\left(x + \frac{b}{2a}\right)^2$

★ the graph between x, y is always a parabola.

★ the co-ordinate of vertex are $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

★ If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.





★ the parabola intersects the y-axis at point $(0, c)$.

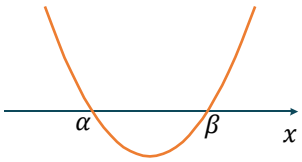
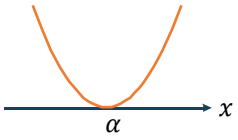
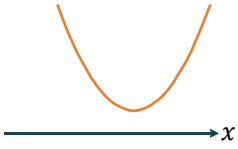
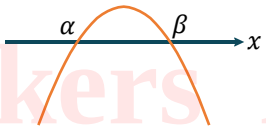
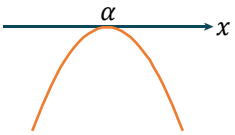
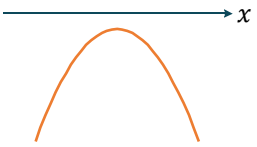
★ the x-co-ordinate of point of intersection of parabola with x-axis are the real roots of the quadratic equation $f(x) = 0$. Hence the parabola may or may not intersect the x-axis.

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Sign of Quadratic Expressions:

The value of expression $f(x) = ax^2 + bx + c$ at $x = x_0$ is equal to y-co-ordinate of the point on parabola $y = ax^2 + bx + c$ whose x-co-ordinate is x_0. Hence if the point lies above the x-axis for some $x = x_0$, then $f(x_0) > 0$ and vice-versa.

We get six different positions of the graph with respect to x-axis as shown.

Graph		Conclusions
(i)		(a) $a > 0$ (b) $D > 0$ (c) Roots are real & distinct. (d) $f(x) > 0$ in $x \in (-\infty, \alpha) \cup (\beta, \infty)$ (e) $f(x) < 0$ in $x \in (\alpha, \beta)$
(ii)		(a) $a > 0$ (b) $D = 0$ (c) Roots are real & equal. (d) $f(x) > 0$ in $x \in \mathbb{R} - \{\alpha\}$
(iii)		(a) $a > 0$ (b) $D < 0$ (c) Roots are imaginary. (d) $f(x) > 0 \forall x \in R$.
(iv)		(a) $a < 0$ (b) $D > 0$ (c) Roots are real & distinct. (d) $f(x) < 0$ in $x \in (-\infty, \alpha) \cup (\beta, \infty)$ (e) $f(x) > 0$ in $x \in (\alpha, \beta)$
(v)		(a) $a < 0$ (b) $D = 0$ (c) Roots are real & equal. (d) $f(x) < 0$ in $x \in \mathbb{R} - \{\alpha\}$
(vi)		(a) $a < 0$ (b) $D < 0$ (c) Roots are imaginary. (d) $f(x) < 0 \forall x \in \mathbb{R}$.

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10. RANGE OF QUADRATIC EXPRESSION $f(x) = ax^2 + bx + c$:

(a) Range:

If $a > 0 \Rightarrow f(x) \in \left[-\frac{D}{4a}, \infty\right)$

If $a < 0 \Rightarrow f(x) \in \left(-\infty, -\frac{D}{4a}\right]$

Hence maximum and minimum values of the expression $f(x)$ is $-\frac{D}{4a}$ in respective cases and it occurs at $x = -\frac{b}{2a}$ (at vertex).

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(b) Range in Restricted Domain:

Given $x \in [x_1, x_2]$

(A) If $-\frac{b}{2a} \notin [x_1, x_2]$ then, $f(x) \in [min\{f(x_1), f(x_2)\}, max\{f(x_1), f(x_2)\}]$

(B) If $-\frac{b}{2a} \in [x_1, x_2]$ then,

$$\Rightarrow f(x) \in \min\left\{f(x_1),\ f(x_2), -\frac{D}{4a}\right\}, \max\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}$$

Illustration 28

If $a < 0$ and $ax^2 + bx + c = 0$ does not have any real roots then prove that

(i) $a - b + c < 0$ (ii) $9a + 3b + c < 0$.

Solution:

$a < 0$ and $D < 0$

$\Rightarrow f(x) = ax^2 + bx + c < 0$ for all $x \in R$

$\Rightarrow f(-1) = a - b + c < 0$ and $f(3) = 9a + 3b + c < 0$

Illustration 29

Find the maximum and minimum values of $f(x) = x^2 - 5x + 6$.

Solution:

minimum of $f(x) = -\left[\frac{D}{4a}\right]$ at $x = -\frac{b}{2a} = -\left(\frac{25-24}{4}\right)$ at $x = \frac{5}{2} = -\frac{1}{4}$

maximum of $f(x) \to \infty$

Hence range is $\left[-\frac{1}{4}, \infty\right)$.

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Illustration 30

Find the range of rational expression $y = \frac{x^2+x+1}{x^2+1}$ if x is real.

Solution:

$y = \frac{x^2+x+1}{x^2+1} \Rightarrow (y-1)x^2 - x + (y-1) = 0$ ...(i)

Case I:

if $y \neq 1$, then equation (i) is quadratic in x

$\therefore D \geq 0 \Rightarrow 1 - 4(y-1)^2 \geq 0$

$\Rightarrow -\frac{1}{2} \leq (y-1) \leq \frac{1}{2} \Rightarrow \frac{1}{2} \leq y \leq \frac{3}{2}$

$y \in \left[\frac{1}{2}, \frac{3}{2}\right] - \{1\}$

Case II:

if $y = 1$, then equation becomes

$-x = 0$

$\Rightarrow x = 0$ which is possible as x is real.

$\therefore$ Range $\left[\frac{1}{2}, \frac{3}{2}\right]$

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Illustration 31

Find the least value of $y = \frac{(6x^2-22x+21)}{5x^2-18x+17}$ for real x ?

Solution:

$y = \frac{(6x^2-22x+21)}{5x^2-18x+17}$

or $(6 - 5y)x^2 - 2x(11 - 9y) + 21 - 17y = 0$

$\Rightarrow D \geq 0 \Rightarrow 4(11 - 9y)^2 - 4(6 - 5y)(21 - 17y) \geq 0$

$\Rightarrow -4y^2 + 9y - 5 \geq 0$

$\Rightarrow 4y^2 - 9y + 5 \leq 0$

$\Rightarrow 4(y - 1)\left(y - \frac{5}{4}\right) \leq 0$

$\Rightarrow y \in \left[1, \frac{5}{4}\right]$

least value is 1.

Illustration 32

Find the values of a for which the expression $\frac{ax^2+3x-4}{3x-4x^2+a}$ assumes all real values for real values of x.

Solution:

Let $y = \frac{ax^2+3x-4}{3x-4x^2+a}$

$\Rightarrow x^2(a + 4y) + 3(1 - y)x - (4 + ay) = 0$

If $x \in R, D \geq 0$

$\Rightarrow 9(1 - y)^2 + 4(a + 4y)(4 + ay) \geq 0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$

for all $y \in R, (9 + 16a) > 0$ & $D \leq 0$

$\Rightarrow (4a^2 + 46)^2 - 4(9 + 16a)(9 + 16a) \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$

$\Rightarrow a^2 - 8a + 7 \leq 0 \Rightarrow 1 \leq a \leq 7$

$\Rightarrow 9 + 16a > 0$ & $1 \leq a \leq 7$

Taking intersection, $a \in [1, 7]$

Now, checking the boundary values of a

For $a = 1$

$\Rightarrow y = \frac{x^2 + 3x - 4}{3x - 4x^2 + 1} = -\frac{(x - 1)(x + 4)}{(x - 1)(4x + 1)}$

$\because x \neq 1 \Rightarrow y \neq -1$

$\Rightarrow a = 1$ is not possible.

If $a = 7$

$\Rightarrow y = \frac{7x^2+3x-4}{3x-4x^2+7} = \frac{(7x-4)(x+1)}{(7-4x)(x+1)}$ $\quad \because x \neq -1 \Rightarrow y \neq -1$

So y will assume all real values for some real values of x.

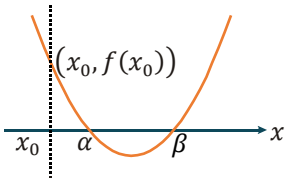
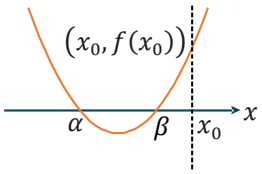
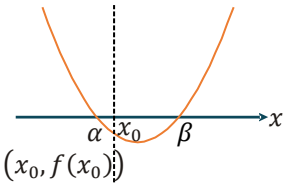
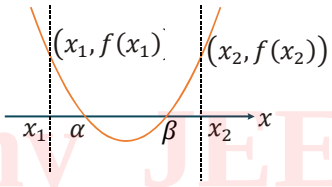
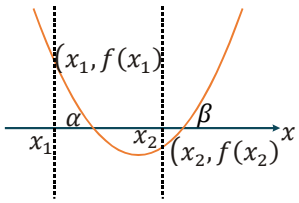
So $a \in (1,7)$

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11. LOCATION OF ROOTS:

Let $f(x) = ax^2 + bx + c$, where $a > 0$ & $a, b, c \in \mathbb{R}$. Let α, β are roots of the equations $f(x) = 0$ and $\alpha < \beta$

Restrictions on roots	Required conditions	Graphical Representation
$\alpha > x_0$ and $\beta > x_0$	$b^2 - 4ac \geq 0$ & $f(x_0) > 0$ & $-\frac{b}{2a} > x_0$	
$\alpha < x_0$ and $\beta < x_0$	$b^2 - 4ac \geq 0$ & $f(x_0) > 0$ & $-\frac{b}{2a} < x_0$	
$\alpha < x_0 < \beta$	$f(x_0) < 0$	
$x_1 < \alpha < \beta < x_2$	$b^2 - 4ac > 0$ & $f(x_1) > 0$ & $f(x_2) > 0$ & $x_1 < -\frac{b}{2a} < x_2$	
exactly one root of $f(x) = 0$ lies in the interval (x_1, x_2) i.e. $x_1 < \alpha < x_2 < \beta$ or $\alpha < x_1 < \beta < x_2$	$f(x_1).f(x_2) < 0$ (check end points)	

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Illustration 33

Let $x^2 - (m-3)x + m = 0 (m \in \mathbb{R})$ be a quadratic equation, then find the values of 'm' for which

(a) both the roots are greater than 2.

(b) both roots are positive.

(c) one root is positive and other is negative.

(d) one root is greater than 2 and other smaller than 1.

(e) roots are equal in magnitude and opposite in sign.

(f) both roots lie in the interval $(1, 2)$.

Solution:

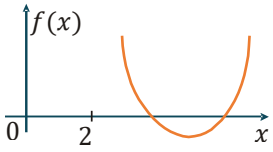
(a) **Condition - I :**

$D \geq 0 \Rightarrow (m-3)^2 - 4m \geq 0$

$\Rightarrow m^2 - 10m + 9 \geq 0$

$\Rightarrow (m-1)(m-9) \geq 0$

$\Rightarrow m \in (-\infty, 1] \cup [9, \infty)$ ...(i)



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Condition - II :

$f(2) > 0 \Rightarrow 4-(m-3)\,2+m>0$

$\Rightarrow m<10$ ...(ii)

Condition - III :

$\Rightarrow -\frac{b}{2a}>2 \Rightarrow \frac{m-3}{2}>2 \Rightarrow m>7$ ...(iii)

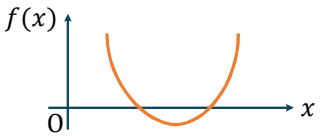
Intersection of (i), (ii) and (iii) gives $m \in [9,10)$

(b) **Condition - I :** $D \geq 0 \quad \Rightarrow m \in (-\infty,1] \cup [9,\infty)$

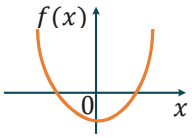
Condition - II : $f(0) > 0 \Rightarrow m>0$

Condition - III : $-\frac{b}{2a} > 0 \Rightarrow \frac{m-3}{2} > 0 \Rightarrow m>3$

intersection gives $m \in [9,\infty)$



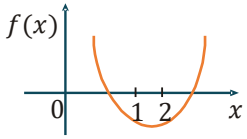
(c) **Condition - I :** $f(0)<0 \Rightarrow m<0$



(d) **Condition - I :** $f(1)<0 \Rightarrow 4<0 \Rightarrow m \in \phi$

Condition - II : $f(2)<0 \Rightarrow m>10$

Intersection gives $m \in \phi$ **Ans.**



(e) sum of roots $= 0 \Rightarrow m=3$

and $f(0)<0 \Rightarrow m<0$

$\therefore m \in \phi$ **Ans.**

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(f) **Condition - I :** $D \geq 0 \quad \Rightarrow m \in (-\infty,1] \cup [9,\infty)$

Condition - II : $f(1)>0 \Rightarrow 1-(m-3)+m>0 \Rightarrow 4>0$ which is true $\forall m \in R$

Condition - III : $f(2)>0 \Rightarrow m<10$

Condition - IV : $1<-\frac{b}{2a}<2 \Rightarrow 1<\frac{m-3}{2}<2 \Rightarrow 5<m<7$

intersection gives $m \in \phi$

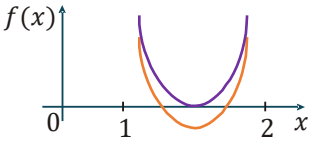
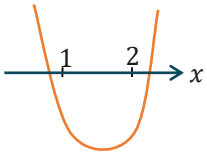


Illustration 34

if α, β are roots of quadratic equation $x^2-2p(x-4)-15=0$ then set of value of p for which one root is less than 1 and other root is greater than 2?

Solution:



This is clear from graph that

$f(1)<0$ and $f(2)<0$

Condition - I:

$f(1)<0 \Rightarrow 1-2p(-3)-15<0$

$6p<14 \Rightarrow p<\frac{7}{3}$ ...(i)

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Condition - II:

$f(2) < 0 \Rightarrow 4 + 4p - 15 < 0$

$4p < 11 \Rightarrow P < \frac{11}{4}$ ...(ii)

by taking intersection of (i) and (ii)

$p \in \left(-\infty, \frac{7}{3}\right)$

Illustration 35

Find value of k for which one root of equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is less than 2.

Solution:

$4 - 2(k+1) + k^2 + k - 8 < 0 \Rightarrow k^2 - k - 6 < 0$

$(k-3)(k+2) < 0 \Rightarrow -2 < k < 3$

Taking intersection, $k \in (-2, 3)$.

Illustration 36

Find all possible values of a for which exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval $(0, 3)$.

Solution:

$f(0).f(3) < 0$

$\Rightarrow 2a(9 - 3(a+1) + 2a) < 0 \Rightarrow 2a(-a+6) < 0$

$\Rightarrow a(a-6) > 0 \Rightarrow a < 0$ or $a > 6$

Checking the extremes.

If $a = 0, x^2 - x = 0$

$x = 0, 1$

$1 \in (0, 3)$

If $a = 6, x^2 - 7x + 12 = 0$

$x = 3, 4$ But both roots $3, 4 \notin (0, 3)$

Hence solution set is

$a \in (-\infty, 0] \cup (6, \infty)$

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12. GENERAL QUADRATIC EXPRESSION IN TWO VARIABLES:

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors if;

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ OR $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

Illustration 37

If $x^2 + 2xy + 2x + my - 3$ have two linear factor then m is equal to -

(A) 6, 2 (B) –6, 2 (C) 6, –2 (D) –6, –2

Ans. (C)

Solution:

Here $a = 1, h = 1, b = 0, g = 1, f = m/2, c = -3$

So $\Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & m/2 \\ 1 & m/2 & -3 \end{vmatrix} = 0$

$\Rightarrow -\frac{m^2}{4} - (-3 - m/2) + m/2 = 0 \Rightarrow -\frac{m^2}{4} + m + 3 = 0$

$\Rightarrow m^2 - 4m - 12 = 0 \Rightarrow m = -2, 6$

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13. THEORY OF EQUATIONS:

If $\alpha_1,\alpha_2,\alpha_3,\ldots\ldots\alpha_v$ are the roots of the equation.

$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n = 0$ where $\alpha_1,\alpha_2,\ldots,\alpha_n$ are all real & $a_0 \neq 0$ then, $\sum\alpha_1 = -\frac{a_1}{a_0}, \sum\alpha_1\alpha_2 = +\frac{a_2}{a_0}, \sum\alpha_1\alpha_2\alpha_3 = -\frac{a_3}{a_0}$,.....,$\alpha_1\alpha_2\alpha_3\ldots\ldots\ldots\alpha_n = (-1)^n\frac{a_n}{a_0}$

Note:

(i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x-\alpha)$ or $(x-\alpha)$ is a factor of $f(x)$ and conversely.

(ii) Every equation of nth degree $(n \geq 1)$ has exactly n roots & if the equation has more than n roots, it is an identity.

(iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. imaginary roots occur in conjugate pairs.

(iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not square of a rational number.

(v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have odd number of real roots (also at least one real root) between 'a' and 'b'.

Illustration 38

If $2x^3 + 3x^2 + 5x + 6 = 0$ has roots α,β,γ then find $\alpha+\beta+\gamma, \alpha\beta+\beta\gamma+\gamma\alpha$ and $\alpha\beta\gamma$.

Solution:

Using relation between roots and coefficients, we get

$\Rightarrow \alpha+\beta+\gamma = -\frac{3}{2}, \alpha\beta+\beta\gamma+\gamma\alpha = \frac{5}{2}, \alpha\beta\gamma = -\frac{6}{2} = -3.$

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Illustration 39

If α,β,γ are the roots of the equation $x^3 + qx + r = 0$ then find the equation whose roots are

(A) $\alpha+\beta, \beta+\gamma, \gamma+\alpha$ (B) $\alpha\beta, \beta\gamma, \gamma\alpha$

Solution:

(A) Given equation $x^3 + qx + r = 0$ having roots α,β and γ.

Sum of roots $= \alpha+\beta+\gamma = 0$

$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = +q$

$\Rightarrow \alpha\beta\gamma = -r$

For equation having roots $\alpha+\beta, \beta+\gamma, \gamma+\alpha$

Sum of roots $= \alpha+\beta+\beta+\gamma+\gamma+\alpha$

$\Rightarrow 2(\alpha+\beta+\gamma) = 0$

$\Rightarrow (\alpha+\beta)(\beta+\gamma)+(\beta+\gamma)(\gamma+\alpha)+(\alpha+\beta)(\gamma+\alpha) = (\alpha+\beta+\gamma)^2 + \alpha\beta+\beta\gamma+\beta\alpha$

$\Rightarrow 0 + q$

$\Rightarrow q$

Product of roots

$\Rightarrow (\alpha+\beta)(\beta+\gamma)(\gamma+\alpha) = (\alpha\beta+\gamma\alpha+\beta^2+\gamma\beta)(\gamma+\alpha)$

$\Rightarrow \alpha\beta\gamma + \alpha^2\beta + \gamma^2\alpha + \gamma\alpha^2 + \beta^2\gamma + \alpha\beta^2 + \gamma^2\beta + \alpha\beta\gamma$

$\Rightarrow -2r + \alpha^2\beta + \alpha^2\gamma + \beta^2\gamma + \beta^2\alpha + \gamma^2\alpha + \gamma^2\beta$

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$\Rightarrow -2r + \alpha^2(-\alpha) + \beta^2(-\beta) + \gamma^2(-\gamma)$

$\Rightarrow -2r - (\alpha^3 + \beta^3 + \gamma^3)$

$\Rightarrow -2r - (3\alpha\beta\gamma)$ $[\alpha + \beta + \gamma = 0;\ \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma]$

$\Rightarrow -2r - (3 \times (-r))$

$\Rightarrow r$

Equation $\Rightarrow x^3 + qx - r = 0$

(B) For equation having roots $\alpha\beta, \beta\gamma, \gamma\alpha$

Sum of roots = $\alpha\beta + \beta\gamma + \gamma\alpha = q$

$\Rightarrow (\alpha\beta)(\beta\gamma) + (\beta\gamma)(\gamma\alpha) + (\gamma\alpha)(\alpha\beta) = \alpha\beta^2\gamma + \beta\gamma^2\alpha + \gamma\alpha^2\beta$

$\Rightarrow \alpha\beta\gamma(\alpha+\beta+\gamma)$

$\Rightarrow -r(0)$

$\Rightarrow 0$

Product $\Rightarrow (\alpha\beta)(\beta\gamma)(\gamma\alpha) = \alpha^2\beta^2\gamma^2 = r^2$

Equation $\Rightarrow x^3 - qx^2 - r^2 = 0$

Illustration 40

If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, find:

(i) $\sum \alpha^3$

(ii) $\alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$

Solution:

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We know that $\alpha + \beta + \gamma = p$

$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = q$

$\Rightarrow \alpha\beta\gamma = r$

(i) $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma + (\alpha + \beta + \gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\}$

$= 3r + p\{p^2 - 3q\} = 3r + p^3 - 3pq$

(ii) $\alpha^2(\beta + \gamma) + \beta^2(\alpha + \gamma) + \gamma^2(\alpha + \beta) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma)$

$= p(\alpha^2 + \beta^2 + \gamma^2) - 3r - p^3 + 3pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$

14. AN SPECIAL APPLICATION ON FINDING THE ROOTS:

It α & β are roots of quadratic equation

$ax^2 + bx + c = 0$ and $S_n = p\alpha^n + q\beta^n$; $p, q \in$ complex number

$a\alpha^2 + b\alpha + c = 0$ & $a\beta^2 + b\beta + c = 0$

Multiply both side by $p\alpha^n$ & $q\beta^n$ in both equation respectively.

$ap\alpha^{n+2} + bp\alpha^{n+1} + cp\alpha^n = 0$ ....(i)

$ap\beta^{n+2} + bp\beta^{n+1} + cp\beta^n = 0$ ....(ii)

(i) + (ii)

$a(p\alpha^{n+2} + q\beta^{n+2}) + b(p\alpha^{n+1} + q\beta^{n+1}) + c(p\alpha^n + q\beta^n) = 0$

$\boldsymbol{aS_{n+2} + bS_{n+1} + cS_n = 0}$

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Idea for Cubic Equation:

If α, β, γ are the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$ and

$S_n = p\alpha^n + q\beta^n + tr^n; p, q, t \in$ complex number

We can write

$a\alpha^3 + b\alpha^2 + c\alpha + d = 0$

$a\beta^3 + b\beta^2 + c\beta + d = 0$

$a\gamma^3 + b\gamma^2 + c\gamma + d = 0$

Multiply both side by $p\alpha^n$, $q\beta^n$ and $t\gamma^n$ in above three equations respectively

$ap\alpha^{n+3} + bp\alpha^{n+2} + cp\alpha^{n+1} + dp\alpha^n = 0$ ...(i)

$ap\beta^{n+3} + bp\beta^{n+2} + cp\beta^{n+1} + dp\beta^n = 0$ ...(ii)

$ap\gamma^{n+3} + bp\gamma^{n+2} + cp\gamma^{n+1} + dp\gamma^n = 0$ ...(iii)

Adding equation (i), (ii) & (iii)

$\alpha(p\alpha^{n+3} + q\beta^{n+3} + t\gamma^{n+3}) + b(p\alpha^{n+2} + q\beta^{n+2} + t\gamma^{n+2}) + c(p\alpha^{n+1} + q\beta^{n+1} + t\gamma^{n+1}) + d(p\alpha^n + q\beta^n + t\gamma^n) = 0$

$aS_{n+3} + bS_{n+2} + cS_{n+1} + dS_n = 0$

Note

We can extend this idea for higher order polynomial in same way also

Illustration 41

If a and b are the roots of equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to _____.

Ans. (51)

Solution:

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$x^2 - 7x - 1 = 0 < \frac{a}{b}$

By newton's theorem

$S_{n+2} - 7S_{n+1} - S_n = 0$

$S_{21} - 7S_{20} - S_{19} = 0$

$S_{20} - 7S_{19} - S_{18} = 0$

$S_{19} - 7S_{18} - S_{17} = 0$

$$\frac{S_{21} + S_{17}}{S_{19}} = \frac{S_{21} + (S_{19} - 7S_{18})}{S_{19}}$$

$$= \frac{S_{21} + S_{19} - 7(S_{20} - 7S_{19})}{S_{19}}$$

$$= \frac{50S_{19} + (S_{21} - 7S_{20})}{S_{19}}$$

$$= 51 \cdot \frac{S_{19}}{S_{19}} = 51$$

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